

Singapore GeAR

Takehiko Nagakura
Han Tu
Wenzhe Peng
takehiko@mit.edu
hantu@mit.edu
pwz@mit.edu
Massachusetts Institute of Technology
Cambridge, Massachusetts, USA

Chili Cheng
Jenny Zhang
Xiaoyun Zhang
Woongki Sung
chilic@mit.edu
jennyz@mit.edu
xiaoyunz@mit.edu
noclew@mit.edu
Massachusetts Institute of Technology
Cambridge, Massachusetts, USA

Katsuyoshi Ishikawa
Shineng Luke Wu
Jia Hui Joey Ong
Nobuo Iwashita
Hisakata Kobayashi
k.ishikawa@kajima.com.sg
s.wu@kajima.com.sg
joey.ong@kajima.com.sg
iwashita@kajima.com.sg
h.kobayashi@kajima.com.sg
Kajima Development Pte Ltd
Singapore

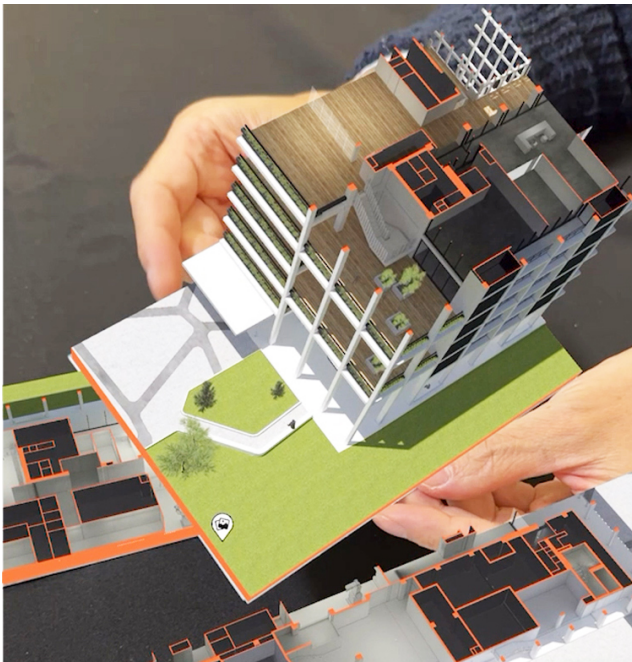


Figure 1: The GEAR.sg is an AR app that shows 3D models of a building on the picture cards of its floor plans.

ABSTRACT

The GEAR.sg App displays 3D models of The GEAR, a recently built research lab in Singapore. Using the smartphone camera, the app presents the building's models and onsite photos on AR picture cards, which illustrate floor plans and allow users to move, flip, and combine the virtual models on them.

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CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality; Interface design prototyping**; • **Applied computing** → **Architecture (buildings)**.

KEYWORDS

Augmented Reality, Architecture, Spatial Design

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1 INTRODUCTION

The GEAR.sg is an interactive smartphone app that introduces The GEAR, a recently built research lab facility in Singapore.

The app design team in Boston worked with the Singapore architects of this building, Kajima Lab for Global Engineering, Architecture & Real Estate, and produced a unique viewer for smartphones, which works as an AR viewer with the presence of the image markers as well as an interactive VR viewer when operated with finger gestures. In a typical use case, four 4x6-inch double-sided picture cards work as the target markers, and each card is printed with a quarter of the first floor plan on one side and the sixth floor plan on the other (Fig 1). Each card in AR view shows the corresponding 3D model displayed on it, and moving and flipping the cards allows viewing of various cross-sectional compositions of the building or the fully assembled building when all cards are combined. Flipping the fifth picture card selects between having the roof on the building or removing it.

2 RELATED PROJECTS

Singapore GeAR project developed the GEAR.sg app from *AR Mail from Harbin*, our earlier project presented in ACM SIGGRAPH 2017 VR Village [Nagakura and Sung 2017], which used a set of four single-sided picture cards to interactively display various compositions of the 3D captured model of a church in China. There, we invented and recognized the powerful basic idea of splitting the building into four quarters, with each corresponding to a part of the plan printed on a picture card that you can simply move by hand. At the same time, we observed shortcomings of its AR design, such as users trying to zoom with pinching finger-gesture in vain, and difficulty representing multi-story buildings. In this project, double-sided picture cards are used to represent a more complex spatial assembly than a single-story space, and its custom UI design demonstrates seamless integration of the AR mode with the VR mode which allows finger gestures to directly manipulate the view. Its development also commenced with a typical BIM (Building Information Modeling) model and onsite photographs provided by the architectural firm instead of scanning an existing building onsite, so that any architect can use the same method of converting their design documentation in BIM and generate a fun, interactive, and information-rich 3D architectural presentation rapidly.

3 APP DESIGN

This app is designed to provide two kinds of user operations as below. It was developed using Unity 3D and Vuforia AR engine.

3.1 AR Markers and Interactions

Nine image markers (illustrating four quarters of the first floor plan, four quarters of the sixth floor plan, and the roof of the building) are registered to this AR app, and each is associated with the corresponding part of the building's 3D model. We found the use of a different type of media displaying the image markers gives the user a distinct interaction experience and engagement (Fig 2).

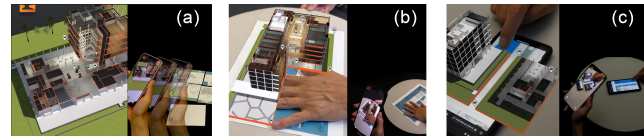


Figure 2: Users viewing 3D models in AR through smartphones compose picture cards (a), cover floor plan sheets by hand (b), or use another phone to display image markers (c).

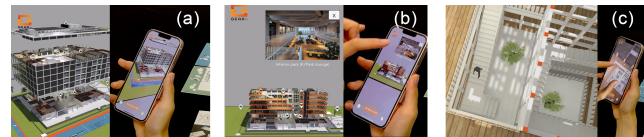


Figure 3: Users can explode the 3D models by double-tapping (a), and bring up the geo-located photos by tapping the flag icons (b). Finger-based VR mode allows zooming by the two-finger pinch action (c), as well as panning and rotation by dragging the view directly on the screen.

By moving and flipping picture cards that have these image markers printed on both sides, various cross-sectional views of the building can be achieved interactively. If the whole plan of either the first or sixth floor is printed on a sheet, hiding a part of the sheet by hand will also reveal a sectional model. The user may also display the image of the floor plans on another smartphone or tablet, and move and scale the image on its screen to observe the transformation of the 3D models emerging in AR.

3.2 Finger-based Manipulation

Touching the smartphone screen initiates the finger-based VR mode, which temporarily suspends the AR markers and allows the user to pan, zoom, and rotate the 3D models directly on the screen by dragging and pinching actions of one or two fingers (Fig 3). Since moving the smartphone does not affect the view in VR mode and the model stays stationary, this mode is more convenient when the user tries to examine details of the 3D model. Also, many users seem to find zooming with two fingers pinching in VR much more intuitive than moving the camera closer to the image marker.

Additional UIs are added to allow double-tapping on the combined 3D models of the building to explode them apart, and clicking on the flag icons embedded in the 3D models to bring up geo-located photos taken in the completed building.

ACKNOWLEDGMENTS

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